



**ALLIANCE
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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

**Title: Importing business datasets from flat files & MySQL and
connecting with any other sources into Power BI**

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[Git hub repo link](#)

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Experiment No.: 1

Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Problem Statement

You are a Data Analyst in a retail company. Sales data is stored in Excel files, customer data in CSV files, and product information in another source. Management wants a centralized reporting system. Import data from multiple sources into Power BI, establish the required connections, and prepare the data for business reporting.

Step 1: Data Collection

- Obtained Sales Data from Excel files (AdventureWorks_Sales_Data_2020.xlsx, 2021.xlsx, 2022.xlsx).
- Obtained Customer Data from CSV files (AdventureWorks_Customer_Lookup.csv).
- Obtained Product Data from another source, i.e., a MySQL database table (Product master table) and supporting CSV lookup files (Product, Product Category, Product Subcategory).

Step 2: Data Import

- Opened Power BI Desktop.
- Imported all datasets using the Get Data option (Get Data → Excel Workbook, Get Data → Text/CSV).
- Connected to the MySQL database using Get Data → Database → MySQL Database, and provided the server name and database name to import the Product table directly from MySQL.
- Loaded Excel, CSV, and MySQL data into Power BI Desktop.

Step 3: Data Cleaning and Transformation

- Removed duplicate records from the Customer and Product tables using Power Query's Remove Duplicates option.
- Handled missing values by replacing blanks/nulls using Replace Values and Fill Down options in Power Query.

- Renamed columns appropriately (e.g., Customer Key to Customer ID, ProductName to Product) for better readability.
- Converted data types where necessary — Date columns to Date type, Key columns to Whole Number, and Price/Cost columns to Fixed Decimal Number.

Step 4: Data Modeling

- Created relationships between the Sales (fact table), Customer, and Product (dimension tables) using common key columns such as Customer Key and Product Key.
- Verified that all relationships were one-to-many, with the lookup tables (Customer, Product) on the “one” side and the Sales table on the “many” side.
- Used the Model view in Power BI Desktop to visually confirm the relationships and cardinality.

Step 5: Dashboard Development

- Created KPI cards for Total Sales, Total Orders, and Total Customers.
- Developed charts for Sales by Category and Sales by Region.
- Added slicers for Month, Category, and Region to enable interactive filtering.
- Designed an interactive business dashboard combining KPI cards, charts, and slicers on a single report page.

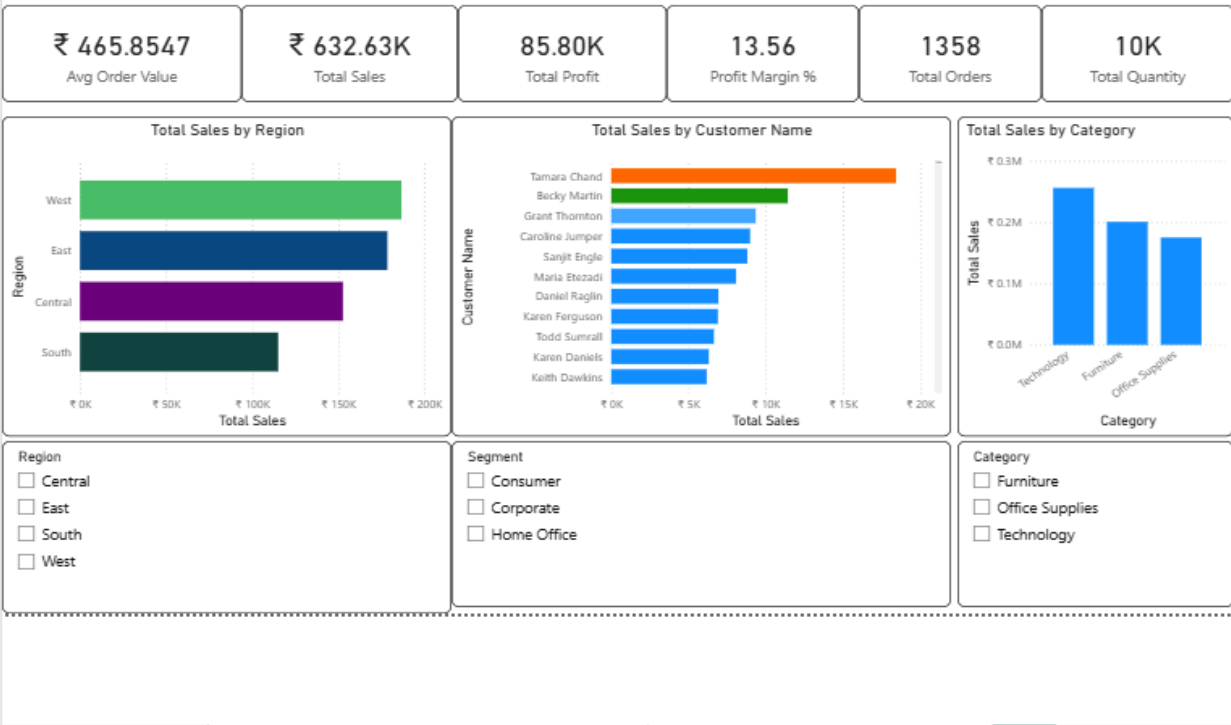
Output

The final Power BI dashboard provides the following outputs:

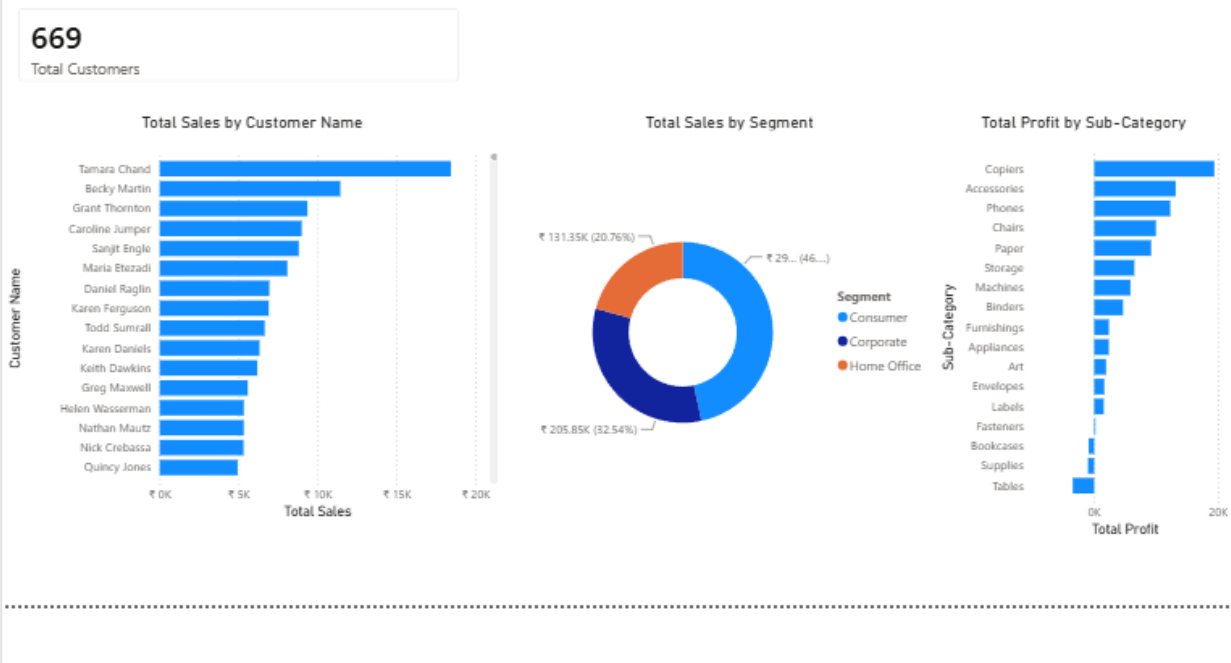
- Total Sales Performance
- Customer Analysis
- Product-wise Sales Analysis
- Monthly Sales Trends
- Interactive Filters and Slicers

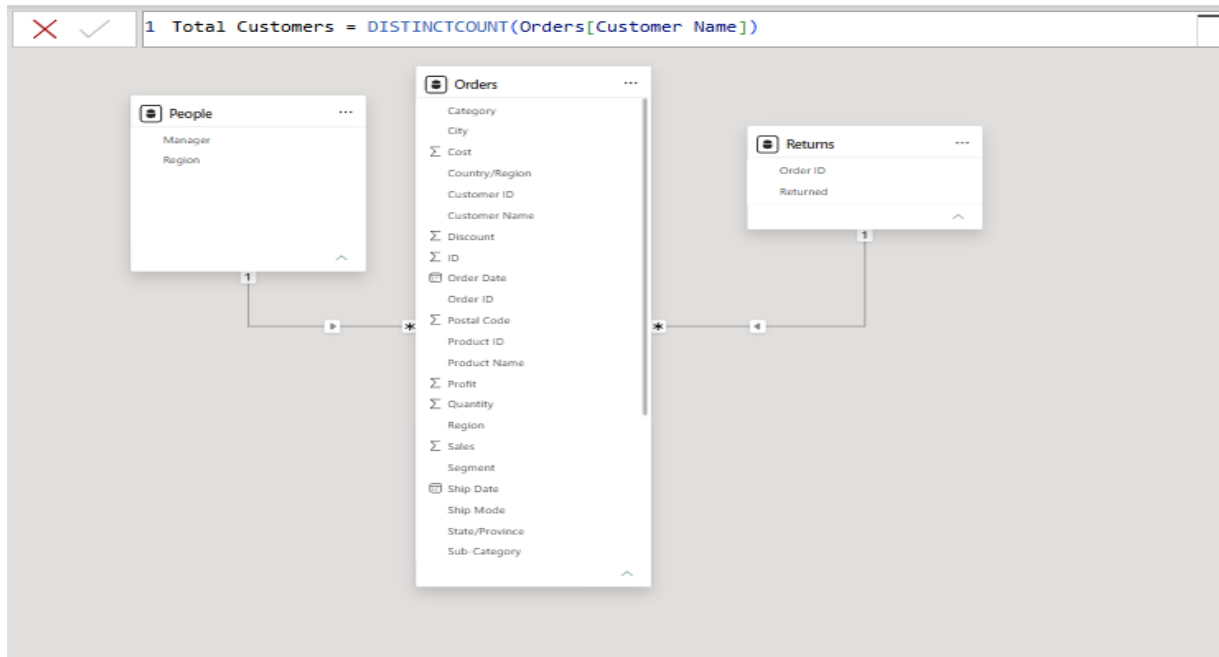
The dashboard enables management to monitor business performance and make data-driven decisions effectively.

Superstore Sales & Profitability Dashboard



Customer & Segment Deep Dive





Conclusion

This project successfully integrated data from multiple sources — Excel, CSV, and MySQL — into Power BI and transformed it into meaningful business insights. Data cleaning, transformation, and modeling techniques were applied to ensure accurate reporting. The developed dashboard provides a centralized view of business performance and supports better decision-making.

Learning Outcomes

Through this project, I learned:

- How to import data from Excel, CSV files, and a MySQL database into Power BI.
- Data cleaning and transformation using Power Query.
- Creating relationships between multiple tables.
- Developing DAX measures and calculated columns.
- Designing interactive dashboards and reports.
- Generating business insights from raw data.

